

KL Divergence and Donsker-Varadhan

Here is the rigorous explanation of the Kullback-Leibler divergence and the Donsker-Varadhan representation, followed by how they are applied in the proof of Theorem 3.2.

1. Kullback-Leibler (KL) Divergence

The Kullback-Leibler divergence is a fundamental measure in information theory and statistics that quantifies how one probability distribution differs from a reference probability distribution.

Definition:

Let P and Q be two probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. Assume P is absolutely continuous with respect to Q (denoted as $P \ll Q$), meaning that for any measurable set A , if $Q(A) = 0$, then $P(A) = 0$. By the Radon-Nikodym theorem, there exists a density function (or likelihood ratio) $\frac{dP}{dQ}$ such that $P(A) = \int_A \frac{dP}{dQ} dQ$.

The KL divergence from Q to P is defined as:

$$D_{\text{KL}}(P\|Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right] = \int_{\mathcal{X}} \log \left(\frac{dP}{dQ} \right) dP$$

If P is not absolutely continuous with respect to Q , then $D_{\text{KL}}(P\|Q) = \infty$.

Key Property:

By expectation Jensen's inequality, $D_{\text{KL}}(P\|Q) \geq 0$, and $D_{\text{KL}}(P\|Q) = 0$ if and only if $P = Q$ almost everywhere. It is not a true metric because it is not symmetric ($D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$) and does not satisfy the triangle inequality.

2. The Donsker-Varadhan Representation

The Donsker-Varadhan representation is a variational formula that expresses the KL divergence as a supremum over a class of functions.

Theorem:

For any two probability measures P and Q on $\mathcal{X}$,

$$D_{\text{KL}}(P\|Q) = \sup_f \left(\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \right)$$

where the supremum is taken over all bounded measurable functions $f : \mathcal{X} \rightarrow \mathbb{R}$ (or any measurable function such that the expectations are well-defined).

Proof:

We must prove two things: (1) that $\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P\|Q)$ for all f , and (2) that the equality is achieved for a specific choice of f .

Step 1: Upper Bound (Weak Duality)

Assume $D_{\text{KL}}(P||Q) < \infty$, which implies $P \ll Q$. Let $p = \frac{dP}{dQ}$ be the Radon-Nikodym derivative.

Consider the quantity $\mathbb{E}_P[f] - D_{\text{KL}}(P||Q)$. We can write this as a single integral with respect to P :

$$\mathbb{E}_P[f] - D_{\text{KL}}(P||Q) = \int_{\mathcal{X}} f dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log(e^f) dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP$$

Since the logarithm function is strictly concave, we can apply Jensen's inequality ($\mathbb{E}[\log(Z)] \leq \log(\mathbb{E}[Z])$):

$$\int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP \leq \log\left(\int_{\mathcal{X}} \frac{e^f}{p} dP\right)$$

By the definition of the Radon-Nikodym derivative, $dP = p dQ$. Substituting this into the right side:

$$\log\left(\int_{\mathcal{X}} \frac{e^f}{p} p dQ\right) = \log\left(\int_{\mathcal{X}} e^f dQ\right) = \log \mathbb{E}_Q[e^f]$$

We have established:

$$\mathbb{E}_P[f] - D_{\text{KL}}(P||Q) \leq \log \mathbb{E}_Q[e^f]$$

Rearranging the terms yields:

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P||Q)$$

Step 2: Achieving the Supremum

To show that the supremum equals $D_{\text{KL}}(P||Q)$, we identify the function f^* that satisfies the equality condition for Jensen's inequality. Jensen's inequality $\mathbb{E}[\log(Z)] \leq \log(\mathbb{E}[Z])$ holds with equality if and only if Z is a constant almost surely.

In our proof, the random variable is $Z = \frac{e^f}{p}$. Setting this to a constant (let's choose 1) gives:

$$\frac{e^{f^*}}{p} = 1 \implies e^{f^*} = p \implies f^* = \log p = \log\left(\frac{dP}{dQ}\right)$$

Substitute this optimal function f^* back into the objective:

$$\mathbb{E}_P[f^*] - \log \mathbb{E}_Q[e^{f^*}] = \int_{\mathcal{X}} \log(p) dP - \log\left(\int_{\mathcal{X}} p dQ\right)$$

Since $p dQ = dP$ and P is a probability measure, $\int_{\mathcal{X}} p dQ = \int_{\mathcal{X}} dP = 1$. Therefore, $\log(1) = 0$, and the expression evaluates exactly to:

$$\int_{\mathcal{X}} \log(p) dP - 0 = D_{\text{KL}}(P||Q)$$

Thus, the supremum is obtained, completing the proof.

3. Application to Theorem 3.2

In the proof of Theorem 3.2 (page 19, Equation 21), the authors are minimizing the following objective with respect to the score function h :

$$\min_h \left(\mathbb{E}_0[h(Y)] + \log \mathbb{E}_{1,P}[e^{-h(Y)}] \right)$$

where $\mathbb{E}_0$ denotes the expectation under the null distribution μ_0 , and $\mathbb{E}_{1,P}$ denotes the expectation under the alternative distribution $\mu_{1,P}$.

To apply the Donsker-Varadhan representation, use the substitution $f = -h$. The minimization problem over h becomes a maximization problem over f :

$$\begin{aligned} \min_h \left(\mathbb{E}_0[h(Y)] + \log \mathbb{E}_{1,P}[e^{-h(Y)}] \right) &= \min_f \left(\mathbb{E}_0[-f(Y)] + \log \mathbb{E}_{1,P}[e^{f(Y)}] \right) \\ &= -\max_f \left(\mathbb{E}_0[f(Y)] - \log \mathbb{E}_{1,P}[e^{f(Y)}] \right) \end{aligned}$$

By the Donsker-Varadhan representation (with $P = \mu_0$ and $Q = \mu_{1,P}$), the term inside the maximum is exactly the definition of the KL divergence:

$$-\max_f \left(\mathbb{E}_{\mu_0}[f] - \log \mathbb{E}_{\mu_{1,P}}[e^f] \right) = -D_{\text{KL}}(\mu_0 \parallel \mu_{1,P})$$

This proves the equality in Equation (21) of the text.

Furthermore, as derived in Step 2 above, the optimal function f^* that achieves this equality is the log-likelihood ratio $f^* = \log \left(\frac{d\mu_0}{d\mu_{1,P}} \right)$. Since $h = -f$, the optimal score function h^* is:

$$h^* = -\log \left(\frac{d\mu_0}{d\mu_{1,P}} \right) = \log \left(\frac{d\mu_{1,P}}{d\mu_0} \right)$$

This aligns precisely with the paper's definition of the optimal score function $h_{\text{gum},\Delta}^*(r)$ stated right below equation (21).

Here is the rigorous proof of why $D_{\text{KL}}(P \parallel Q) \geq 0$ and why equality holds if and only if $P = Q$.

This result relies directly on Jensen's inequality.

1. Statement of Jensen's Inequality

Let X be an integrable random variable defined on a probability space, and let ϕ be a convex function. Jensen's inequality states that:

$$\mathbb{E}[\phi(X)] \geq \phi(\mathbb{E}[X])$$

Furthermore, if ϕ is a **strictly convex** function, the equality $\mathbb{E}[\phi(X)] = \phi(\mathbb{E}[X])$ holds if and only if X is a constant almost surely (i.e., $X = c$ with probability 1, for some constant c).

2. Proof that $D_{\text{KL}}(P \parallel Q) \geq 0$

By definition, the KL divergence is the expected value of the log-likelihood ratio under the distribution P :

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right]$$

To apply Jensen's inequality, we rewrite the term inside the expectation by pulling out a negative sign and taking the reciprocal of the ratio:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P \left[-\log \frac{dQ}{dP} \right]$$

Now, let our random variable be $X = \frac{dQ}{dP}$, and our function be $\phi(t) = -\log(t)$.

The second derivative of $\phi(t)$ is $\phi''(t) = \frac{1}{t^2} > 0$ for all $t > 0$. Because the second derivative is strictly positive, the function $\phi(t) = -\log(t)$ is strictly convex.

Applying Jensen's inequality to this expectation yields:

$$\mathbb{E}_P \left[-\log \left(\frac{dQ}{dP} \right) \right] \geq -\log \left(\mathbb{E}_P \left[\frac{dQ}{dP} \right] \right)$$

Next, we evaluate the expectation on the right side. The expectation of $\frac{dQ}{dP}$ under the distribution P is exactly the integral of dQ over the support of P :

$$\mathbb{E}_P \left[\frac{dQ}{dP} \right] = \int_{\mathcal{X}} \frac{dQ}{dP} dP = \int_{\mathcal{X}} dQ = 1$$

(Note: $\int dQ = 1$ because Q is a valid probability measure).

Substitute this back into the right side of the inequality:

$$-\log(1) = 0$$

Therefore, we conclude:

$$D_{\text{KL}}(P||Q) \geq 0$$

3. Proof of the Equality Condition ($D_{\text{KL}} = 0 \iff P = Q$)

Because the function $\phi(t) = -\log(t)$ is **strictly** convex, the equality in Jensen's inequality holds if and only if the random variable $X = \frac{dQ}{dP}$ is a constant almost everywhere with respect to P . <https://www.math.mcgill.ca/dstephens/556-2014/Handouts/Math556-05-Inequalities.pdf>

Direction 1: If $D_{\text{KL}}(P||Q) = 0$, then $P = Q$

If the divergence is exactly zero, the equality condition of Jensen's inequality must be met. This means:

$$\frac{dQ}{dP} = c \quad (P\text{-almost everywhere})$$

for some constant c .

We already established that $\mathbb{E}_P \left[\frac{dQ}{dP} \right] = 1$. Substituting c into this expectation gives:

$$\mathbb{E}_P[c] = 1 \implies c = 1$$

Thus, $\frac{dQ}{dP} = 1$ almost everywhere. This immediately implies that $dP = dQ$ almost everywhere, meaning the two probability measures are identical: $P = Q$.

Direction 2: If $P = Q$, then $D_{\text{KL}}(P||Q) = 0$

If $P = Q$ almost everywhere, then the Radon-Nikodym derivative $\frac{dP}{dQ} = 1$ almost everywhere. Substituting this directly into the definition of KL divergence yields:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P [\log(1)] = \mathbb{E}_P[0] = 0$$

This completes the proof.